C Program for Polynomial Addition Using Singly Linked List:

```
#include <stdio.h>

#include <stdlib.h>


// Define the structure of a node

struct Node {

   int coefficient;    // Coefficient of the term

   int exponent;       // Exponent of the term

   struct Node* next;  // Pointer to the next term

};


// Function to create a new node

struct Node* createNode(int coefficient, int exponent) {

   struct Node* newNode = (struct Node*)malloc(sizeof(struct Node));

   newNode->coefficient = coefficient;

   newNode->exponent = exponent;

   newNode->next = NULL;

   return newNode;

}


// Function to insert a new node in the polynomial

void insertNode(struct Node** poly, int coefficient, int exponent) {

   struct Node* newNode = createNode(coefficient, exponent);

   if (*poly == NULL) {

      *poly = newNode;  // Insert as the first node if the list is empty

      return;

   }


   struct Node* temp = *poly;

   while (temp->next != NULL) {

      temp = temp->next;  // Traverse to the end of the list
```

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    }
    temp->next = newNode;  // Insert at the end
}

// Function to add two polynomials
struct Node* addPolynomials(struct Node* poly1, struct Node* poly2) {
    struct Node* result = NULL;

    // Traverse both polynomials
    while (poly1 != NULL && poly2 != NULL) {
        if (poly1->exponent > poly2->exponent) {
            insertNode(&result, poly1->coefficient, poly1->exponent);
            poly1 = poly1->next;
        } else if (poly1->exponent < poly2->exponent) {
            insertNode(&result, poly2->coefficient, poly2->exponent);
            poly2 = poly2->next;
        } else {
            // Same exponents, add coefficients
            int sum = poly1->coefficient + poly2->coefficient;
            if (sum != 0) {
                insertNode(&result, sum, poly1->exponent);
            }
            poly1 = poly1->next;
            poly2 = poly2->next;
        }
    }

    // Add remaining terms from poly1
    while (poly1 != NULL) {
        insertNode(&result, poly1->coefficient, poly1->exponent);
        poly1 = poly1->next;
```

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    }

    // Add remaining terms from poly2
    while (poly2 != NULL) {
        insertNode(&result, poly2->coefficient, poly2->exponent);
        poly2 = poly2->next;
    }

    return result;
}

// Function to print a polynomial
void printPolynomial(struct Node* poly) {
    while (poly != NULL) {
        printf("%dx^%d", poly->coefficient, poly->exponent);
        poly = poly->next;
        if (poly != NULL)
            printf(" + ");
    }
    printf("\n");
}

int main() {
    struct Node* poly1 = NULL;
    struct Node* poly2 = NULL;
    struct Node* result = NULL;

    // First Polynomial: 3x^3 + 5x^2 + 6
    insertNode(&poly1, 3, 3);
    insertNode(&poly1, 5, 2);
    insertNode(&poly1, 6, 0);
```

```
    // Second Polynomial: 4x^2 + 2x + 3
    insertNode(&poly2, 4, 2);
    insertNode(&poly2, 2, 1);
    insertNode(&poly2, 3, 0);

    // Display the polynomials
    printf("First Polynomial: ");
    printPolynomial(poly1);

    printf("Second Polynomial: ");
    printPolynomial(poly2);

    // Add the two polynomials
    result = addPolynomials(poly1, poly2);

    // Display the result
    printf("Sum of Polynomials: ");
    printPolynomial(result);

    return 0;
}
```

Output

```
First Polynomial: 3x^3 + 5x^2 + 6x^0
Second Polynomial: 4x^2 + 2x^1 + 3x^0
Sum of Polynomials: 3x^3 + 9x^2 + 2x^1 + 9x^0


=== Code Execution Successful ===
```